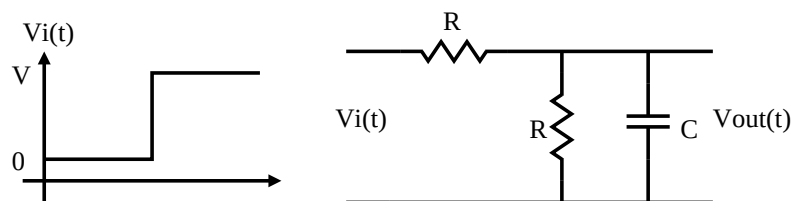


Analog Guide

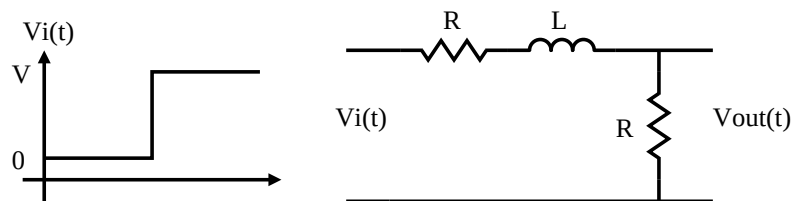
Raja Reddy P, Indian Institute of Science

July 26, 2007

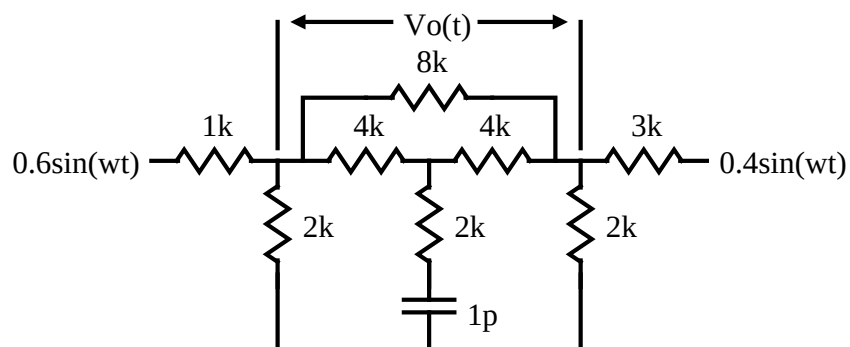
1. Calculate $V_{out}(s)/V_{in}(s)$. plot $V_o(t)$. calculate time constant and pole frequency.



2. In the following circuit plot $i_L(t)$, $V_L(t)$

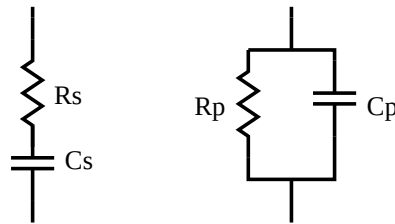


3. Find $V_o(t)$ in the circuit below.

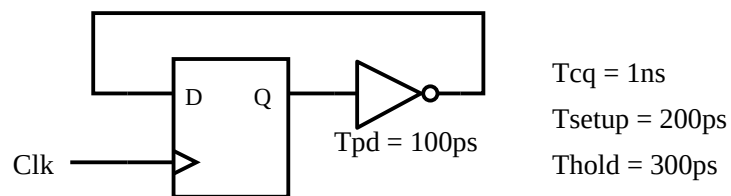


4. A system has an SNR of 60 dB. If an uncorrelated noise of 1 mV is added in a 1V of signal to it, then what is the SNR?

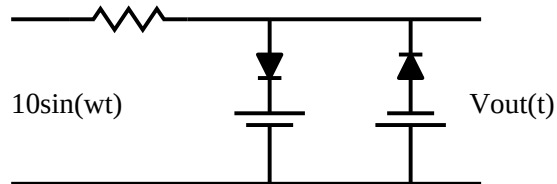
5. Both ckts are equivalent. Express R_p and C_p in terms of R_s and C_s . Find ω range for which these both are equivalent. Assume high quality factor.



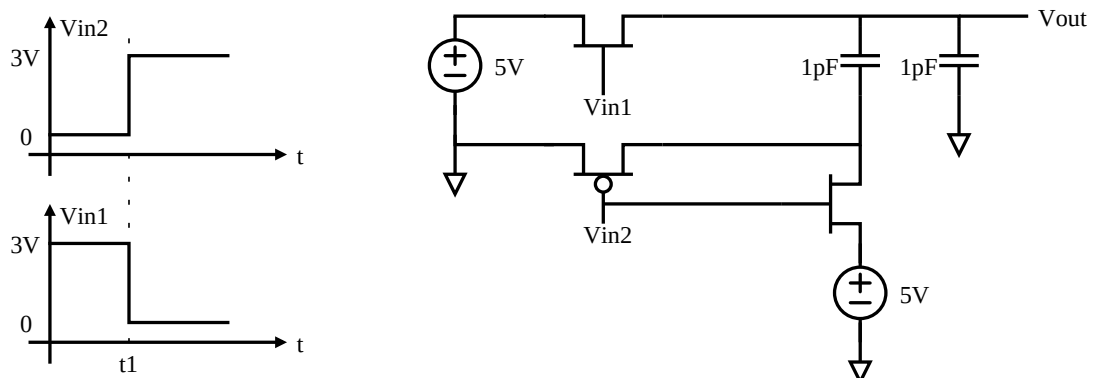
6. Calculate the maximum clock frequency.



7. Plot V_{out} .

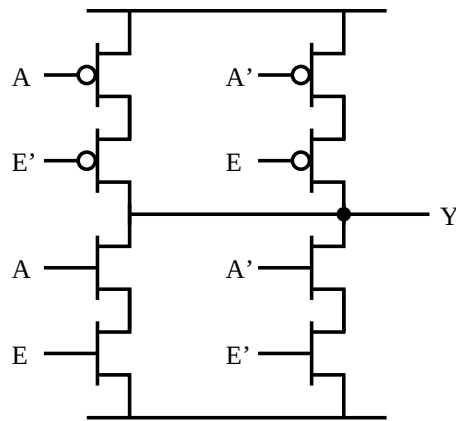


8. Plot V_{out} with respect to the given input waveforms.

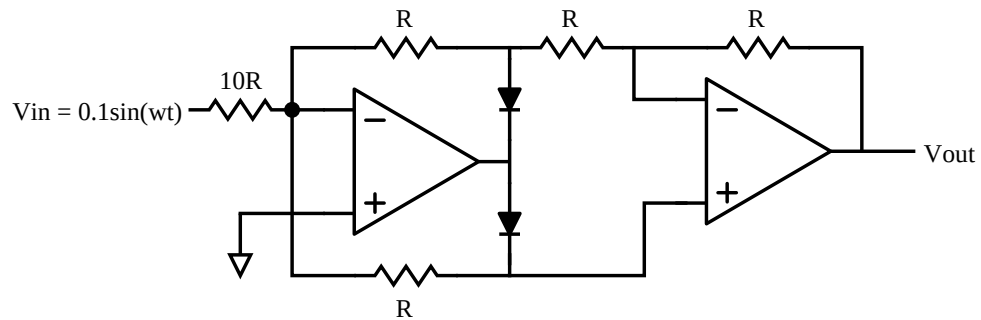


9. Design a divide-by-3 counter using D-flipflops. The duty cycle of the divided clock should be $2/3$.

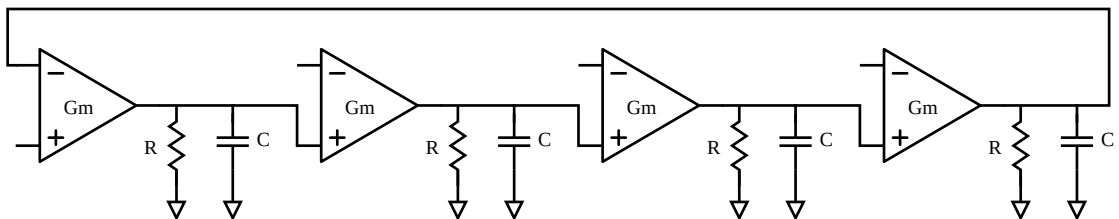
10. What is the function of the following circuit?



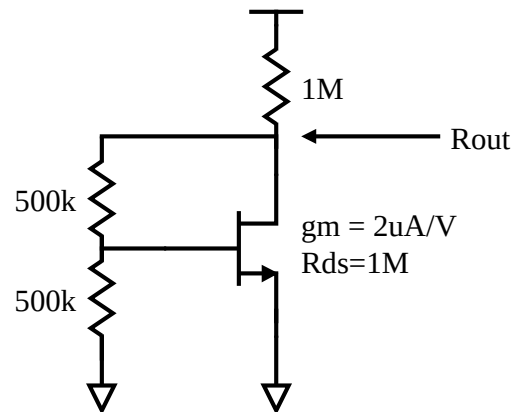
11. Plot the output V_{out} .



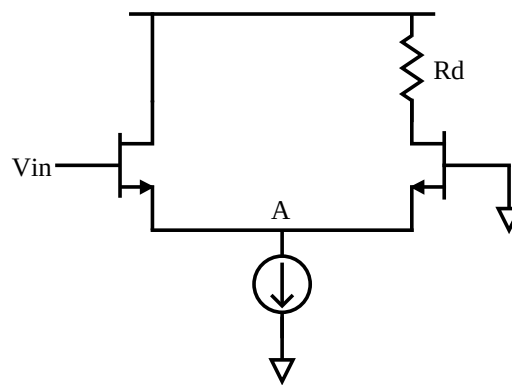
12. Calculate the frequency of oscillation. What is the minimum required G_m for oscillation?



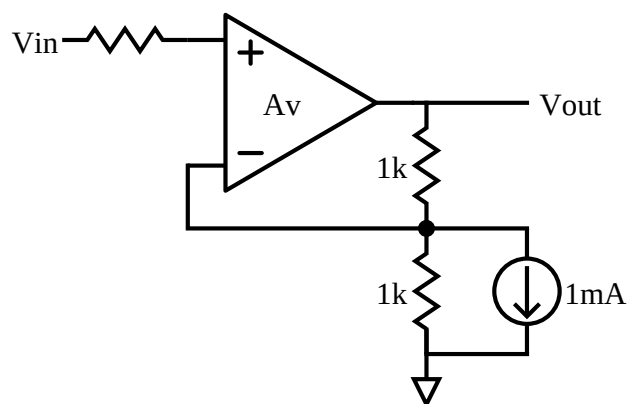
13. Calculate the output impedance.



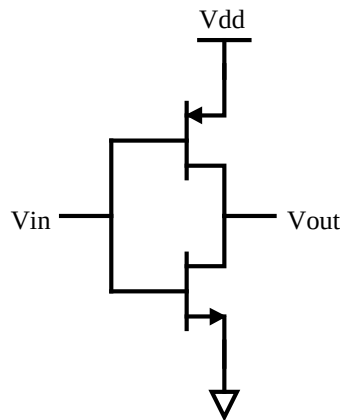
14. Both transistors are biased in saturation. Calculate “ V_A/V_{in} ”. Neglect the body effect.



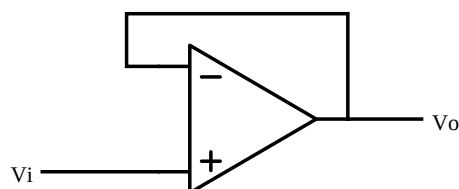
15. Calculate the output if (i) gain $A_v = \text{infinity}$ (ii) gain $A_v = 10$



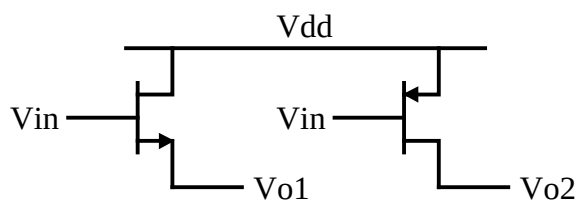
16. If the following inverter biased in the middle of V_{dd} , what is the small signal gain?
(Answer: $g_m r_o$)



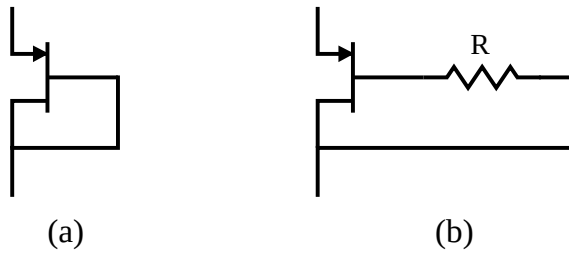
17. Cross-section diagram of the inverter (be able to draw the contact of power supply and ground)
18. From the cross-section of the diagram, be able to draw the parasitic BJT leads to latch-up.
19. How to prevent latch-up (do not forget guard ring, clamping circuits!)
20. Draw the layout of an inverter or NOR/NAND gate.
21. For the following source follower, what is its -3dB bandwidth? How about its stability?



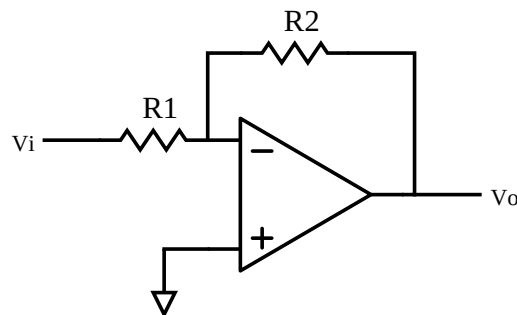
22. For the following circuits, $V_{dd}=5V$, tell me what are V_{o1} and V_{o2} when V_{in} is 5V, 3V, 2.5V and 0V.



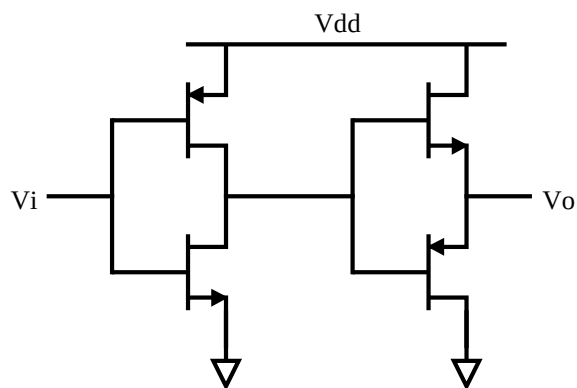
23. What are the effective resistance from source to drain of the following two transistors? (The value of the resistance is R). Answer: both of them are $1/g_m$.



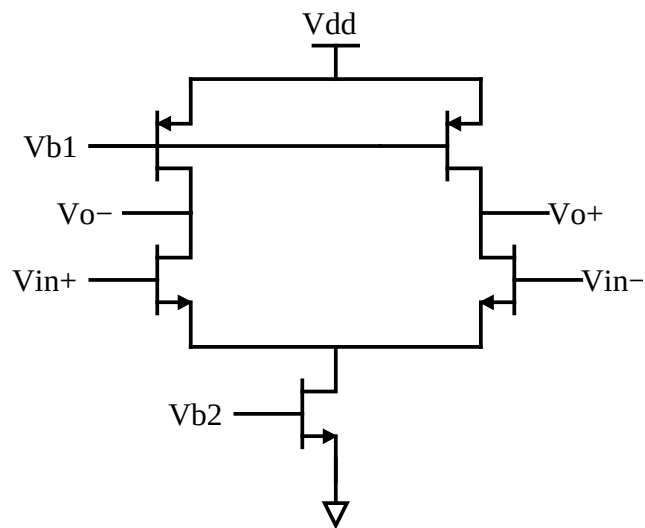
24. In the following figure, if the two resistors are equal, what is its -3dB bandwidth? Compare its stability with that of a source follower.



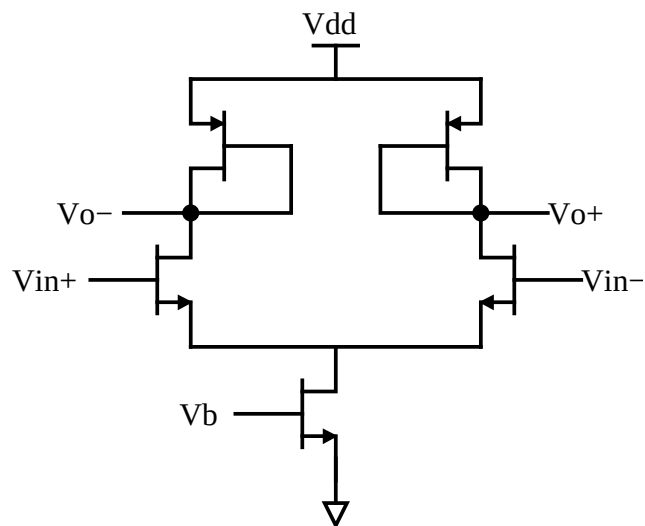
25. For the following circuit, if the input is a rail-to-rail square wave, plot the wave after the inverter and v_o .



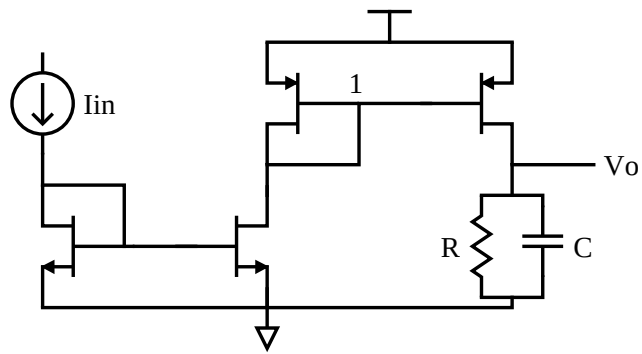
26. For the following circuits, What is the gain? Using what technology to improve the matching of the input transistors? If the bias current increase, what happens to the gain? (Hit: Decrease!!!) What happens to the bandwidth? Replace the NMOS with npn BJT and PMOS with pnp BJT, answer the above questions.(Now gain remains constant with increasing biasing current!)



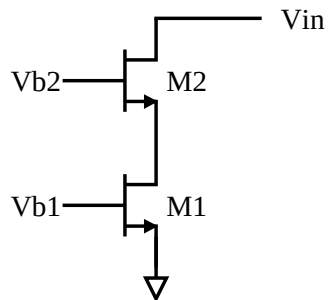
27. For the following circuits, answer the questions again. What are the advantages and disadvantages of these two amplifiers?



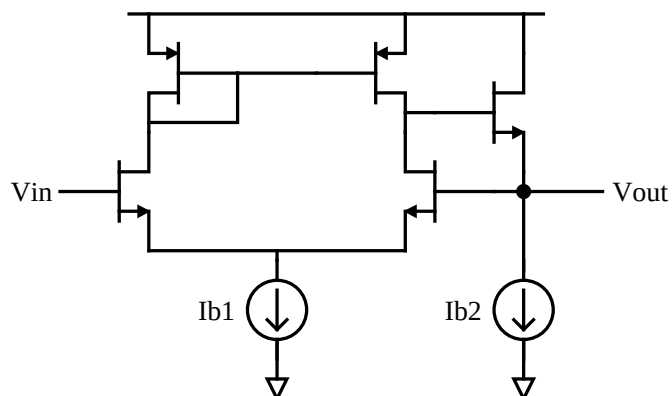
28. You are probing a square wave pulse in the lab that has a rise time of 5 ns and fall time of 2 ns. What is the minimum bandwidth of the oscilloscope to view the signal?
- Answer: The time that it takes an RC circuit to go from 10% to 90% of its final value is $t = \ln 9 \cdot RC$. The bandwidth of the oscilloscope larger than $BW = \ln 9 / (2 \cdot \pi \cdot 2 \text{ ns}) = 174 \text{ MHz}$. Choose a 200 MHz or faster oscilloscope. To reduce error, choose an oscilloscope 3 times faster than the calculated value, or 600 MHz.
29. What is the low frequency gain of the following circuits? The input is the input current I_{in} . Where does the dominant pole locate? How about the pole at node 1?



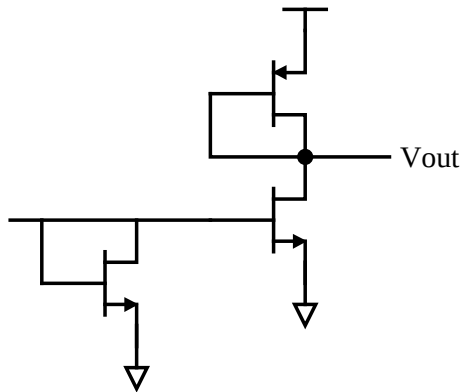
30. For the following circuit, the threshold voltage of the transistor is 0.7V. $V_{b1}=1\text{V}$, $V_{b2}=2\text{V}$. When V_{in} change from 5V to 0V, draw the current flow through the transistors VS V_{in} . (This question was supplied by Wang Ge)



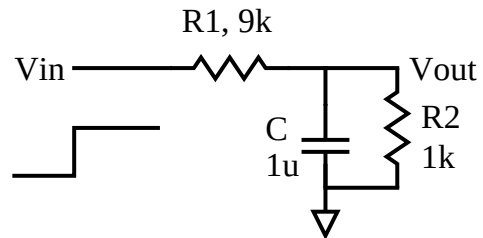
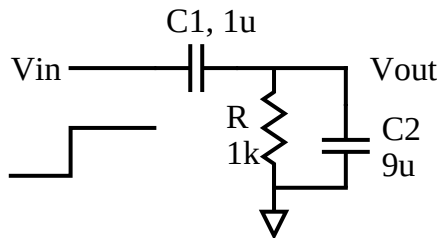
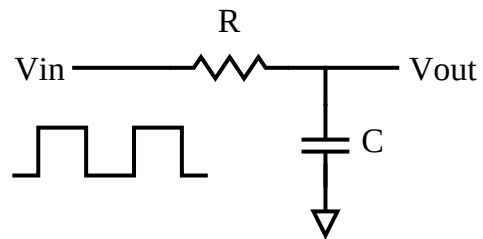
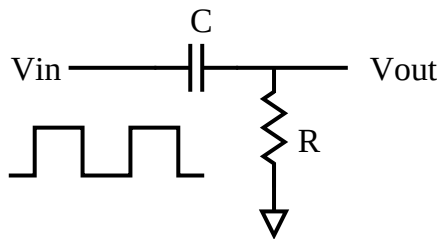
31. For the following circuit, what is the gain of V_{out}/V_{in} ? Where is the Feedback and what is the function of feedback?



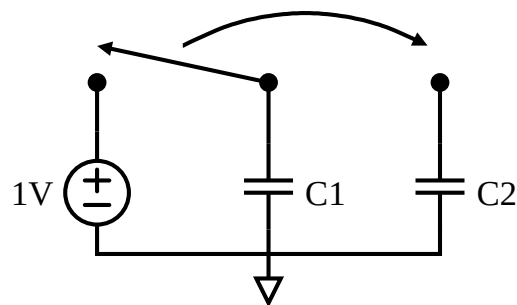
32. For the following circuits, the small signal input is I_{in} , the small signal output is v_{out} , what is the small signal output? What is the gain?



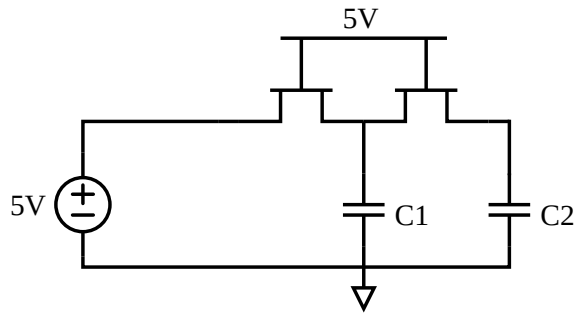
33. Plot the V_{out} wave form of the following circuits:



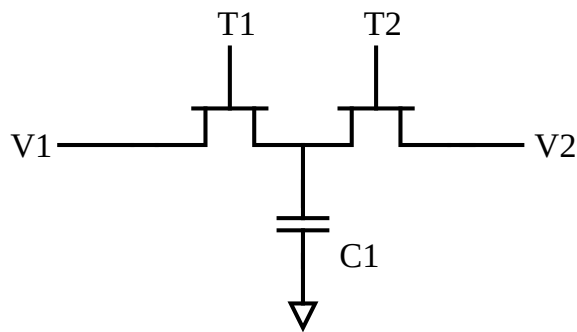
34. For the following circuit, at time 0, the switch switches from A to B, figure out the voltage wave form at B.



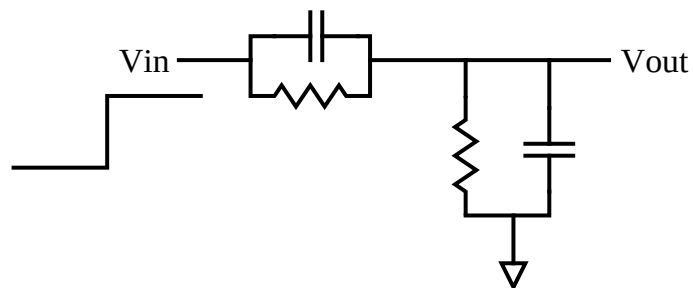
35. For the following circuit, what is the voltage value at A and B? (The V_t of the transistor is 1V).



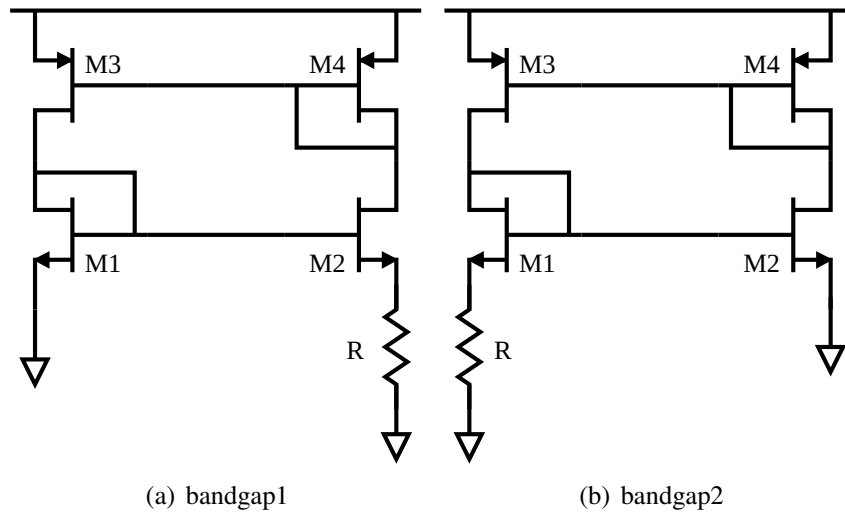
36. The capacitor of the following figure is connected with two ideal MOS switches. Switches T1 and T2 are alternately turned on with a frequency f_{fc} . What is the average current flowing from node 1 to node 2? What is the equivalent impedance from node 1 to node 2?



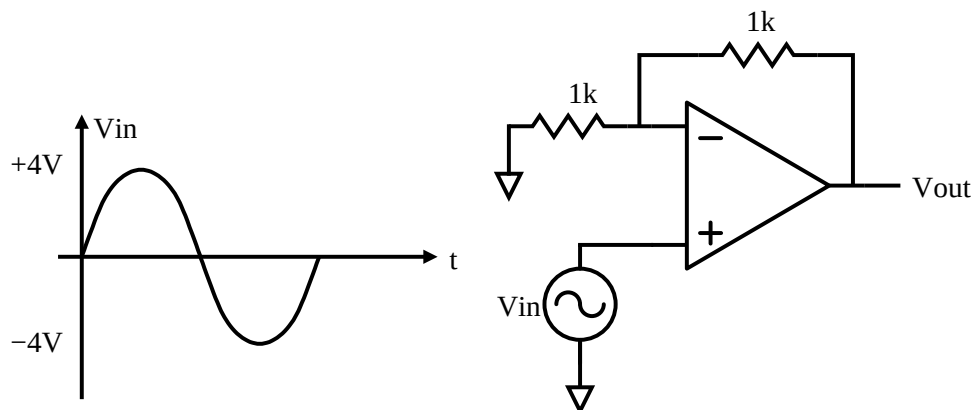
37. Plot the output of the following RC circuit



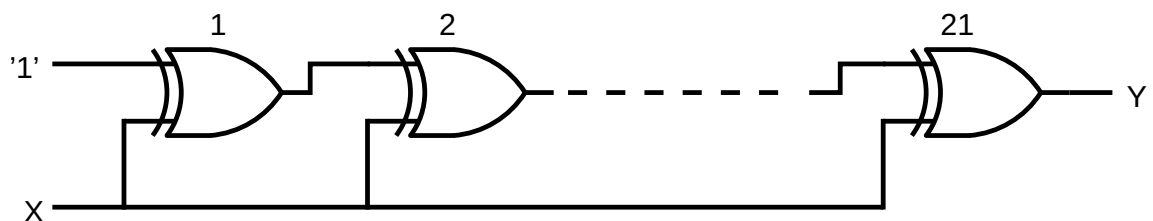
38. How the following two circuits differ in their functionality and which one is a better reference?



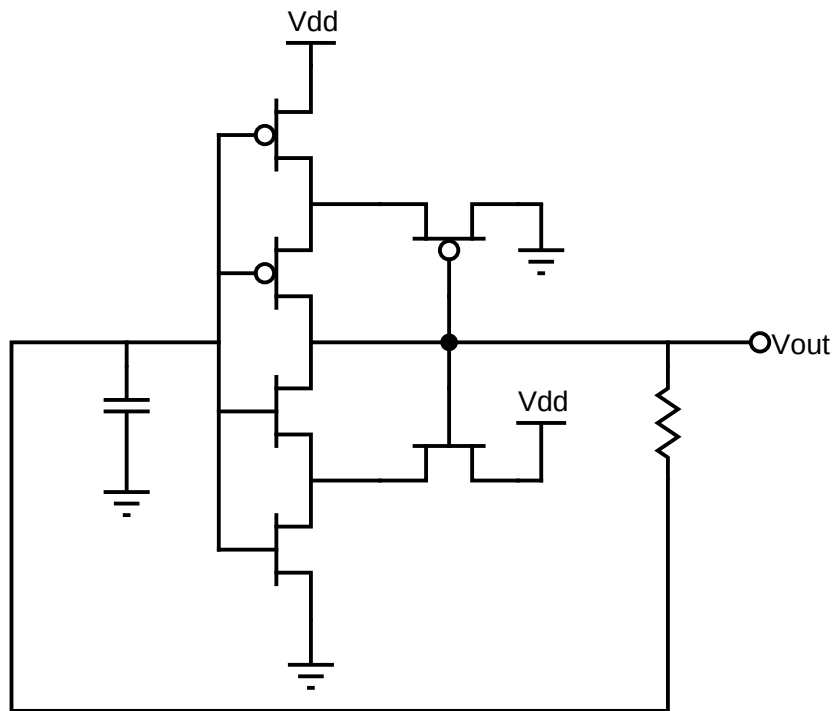
39. Draw V_o . Initially V_o is at +5V. $V_{sat} = 5V$



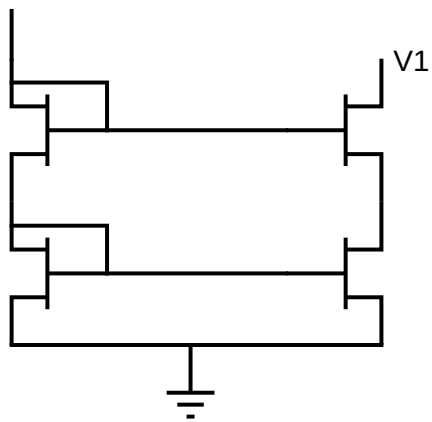
40. what is the output Y in the figure shown below.



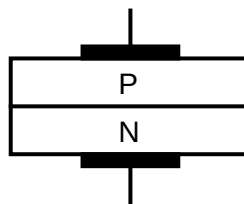
41. Draw the output waveform.



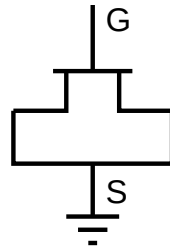
42. what is the minimum value of V_1 .



43. How the diode should be doped for faster electron-hole collection at terminals.



44. In which region the capacitance C_{gs} is maximum.

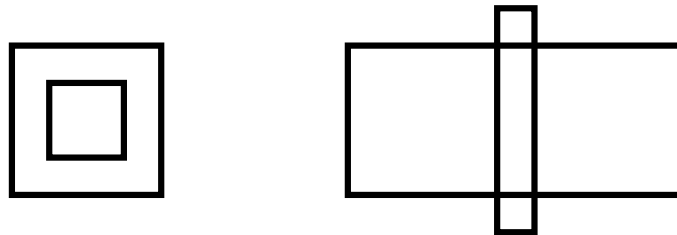


45. How a pn-diode should be doped for maximum capacitance?

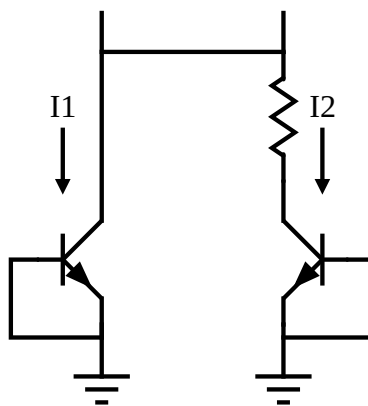
46. How to reduce the crowbar (short circuit current) in an inverter?

47. How the latchup and ESD robustness are affected with increased substrate doping?

48. I have not remembered this question exactly. Two different MOS structures are given and asked to find about the sidewall and bottom capacitance.



49. How I_1 and I_2 changes with temperature

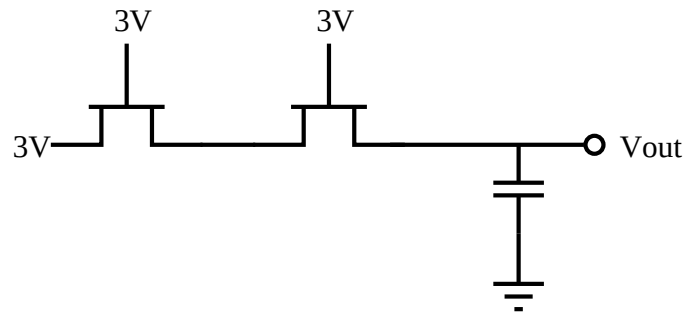


50. Decreased drain doping results in (choose from the options below)

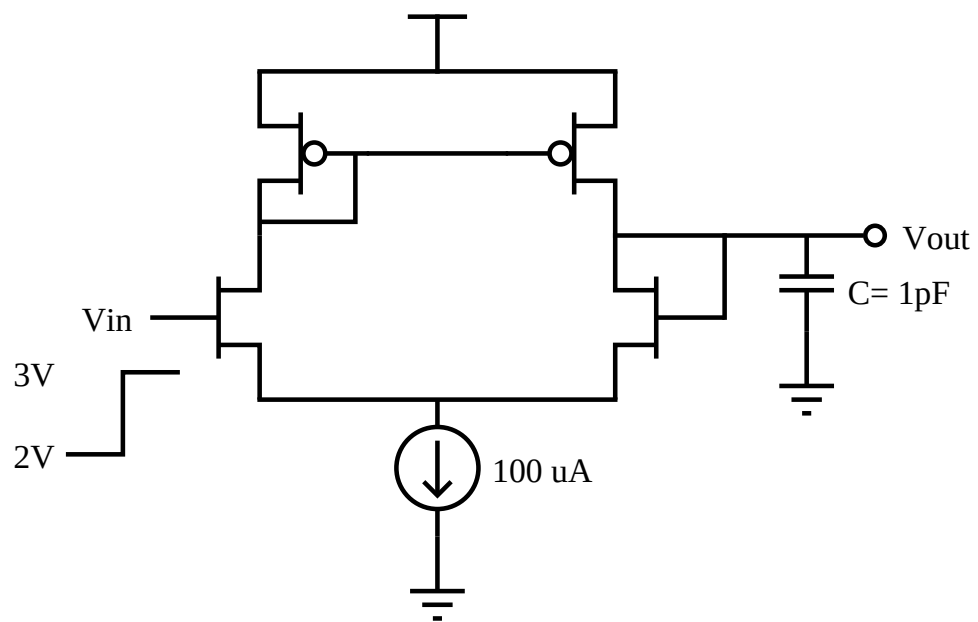
(a) High gate oxide reliability

- (b) Hot carrier reliability
- (c) Decreases drain contact resistance

51. what is the output voltage



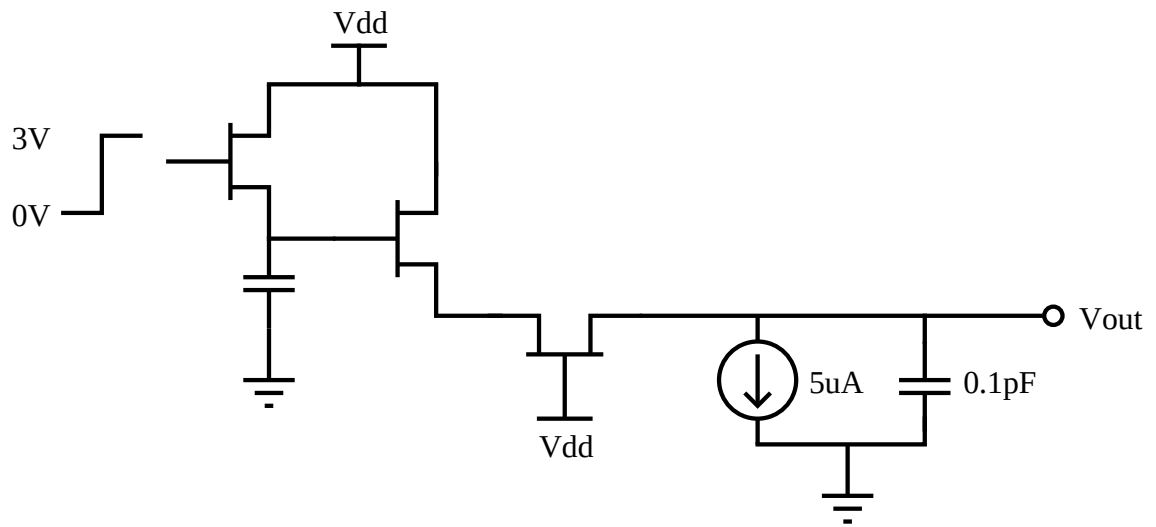
52. what is the slew rate of output voltage



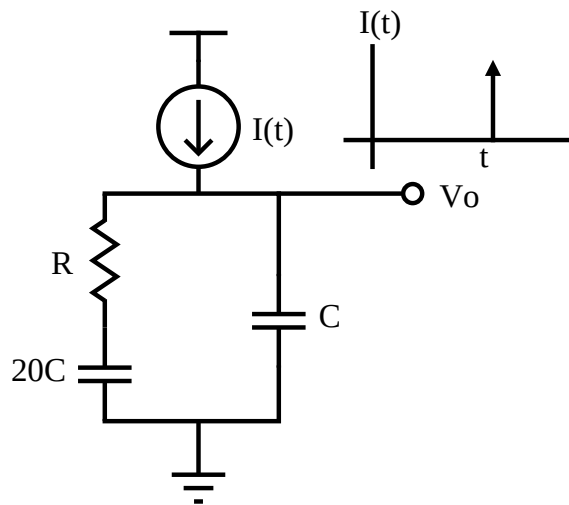
53. For an NMOS in saturation, the drain current depends on (choose from the options below)

- (a) Surface mobility
- (b) Bulk mobility
- (c) Both
- (d) None

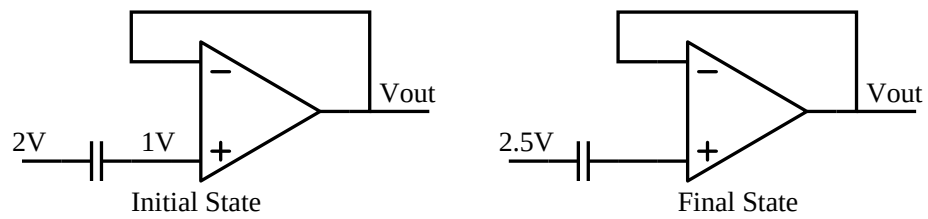
54. what is the output voltage after 5 seconds



55. plot the output voltage



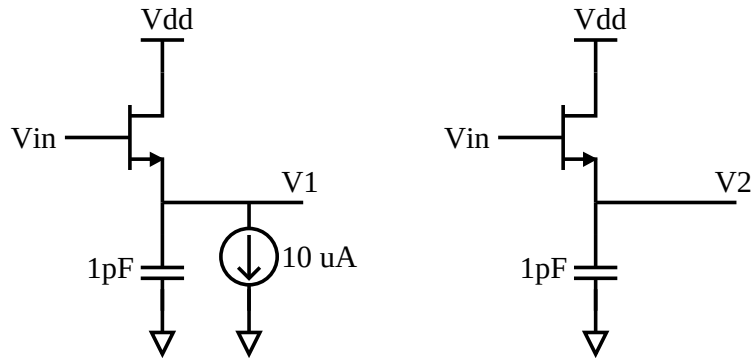
56. what is V_{out} in the final state?



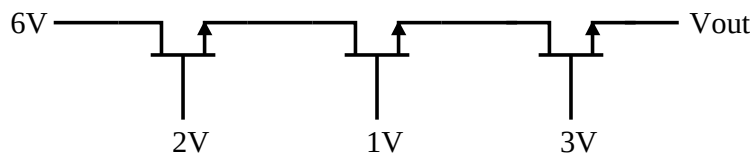
57. Frequency spectrum of ideal squarewave (choose from the options below)

- (a) Only $C_0 = 0$
- (b) Odd terms are zero
- (c) Even terms are zero

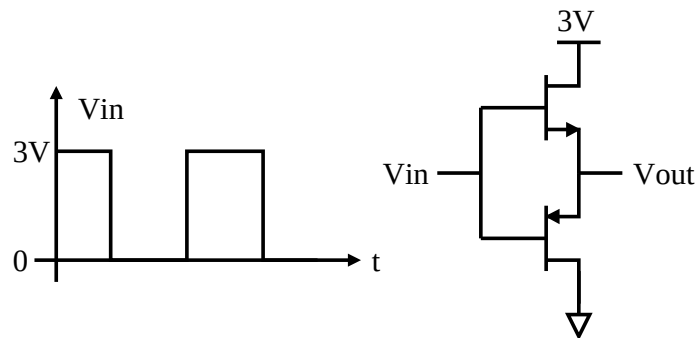
58. Initially $V_1 = V_2 = 1V$ and $V_{in} = 0.7V$. If V_{in} changes to $1.5V$ what are the final states?



59. what is the steady state output voltage, Threshold voltage is $0.5V$



60. Plot the output voltage

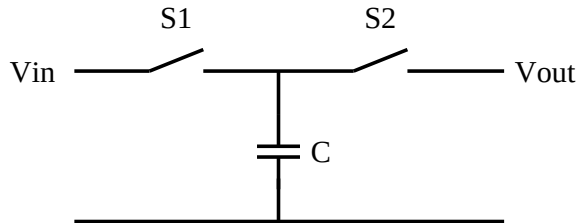


61. If V_{th} of NMOS increases, for an inverter NM_L and NM_H will

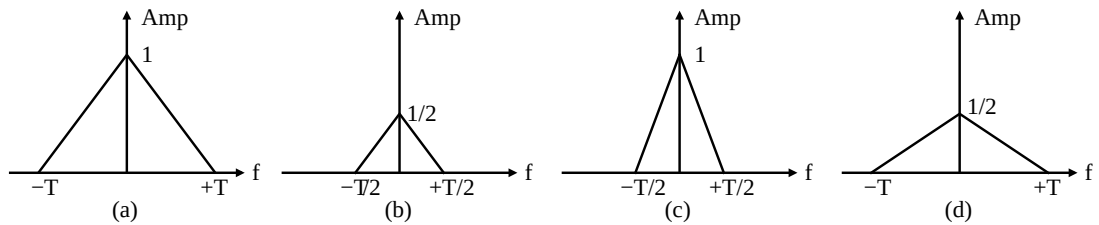
- (a) both increase

- (b) both decrease
- (c) NM_L increase NM_H decrease
- (d) NM_L decrease NM_H increase

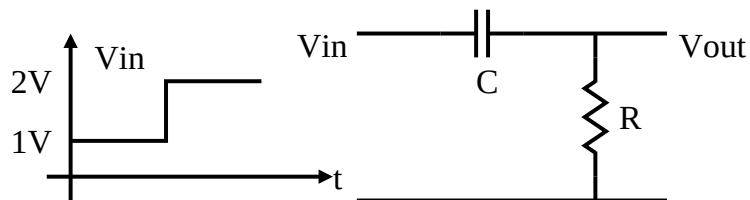
62. What is the equivalent value of resistance.



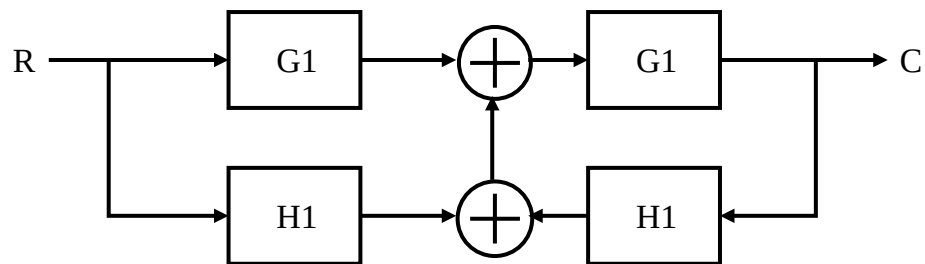
63. Impulse response of interpolation filter is.



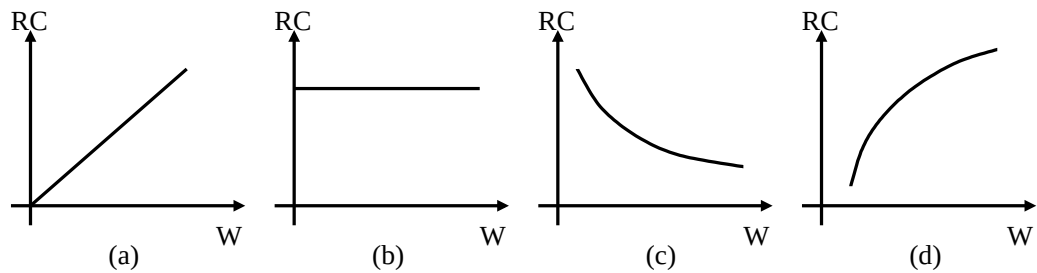
64. plot the output voltage



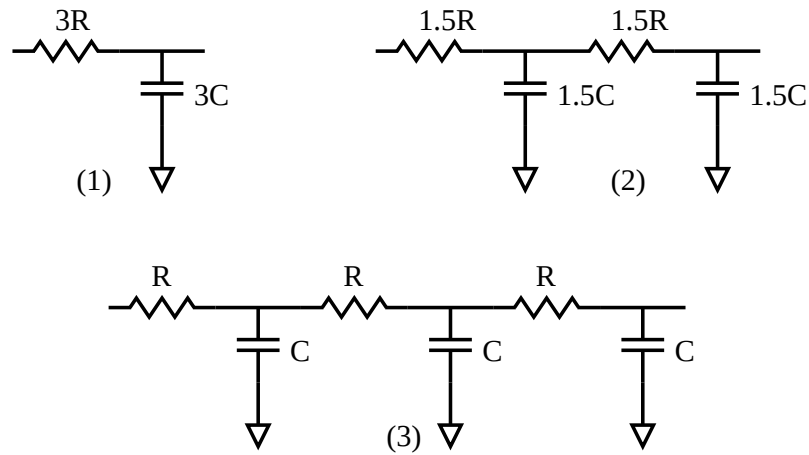
65. What is the transfer function



66. A planar metal is used for an interconnect. Neglecting the fringing capacitance the RC time constant will be



67. Relationship between the time constants of figures 1,2,3



68. In NMOS at weak inversion, the current is dominated by

- (a) drift current
- (b) diffusion current
- (c) both
- (d) none of the above

69. In MOS, channel is inverted when the surface potential reaches

- (a)
- (b)
- (c)
- (d)

70. Band-to-band tunneling occurs when

- (a) both dopings are high
- (b) both dopings are low
- (c) one is high and other is low
- (d) independent of doping

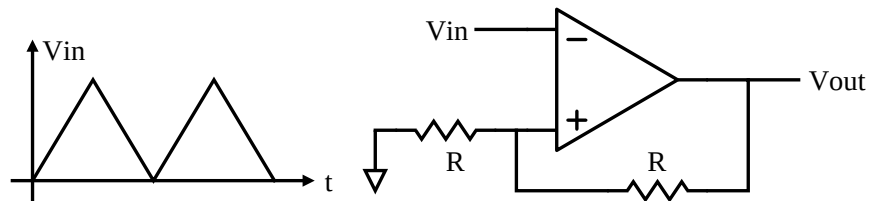
71. One question on $np = n_i^2$

72. Logic expression $[(a + b')(a' + b)]'$

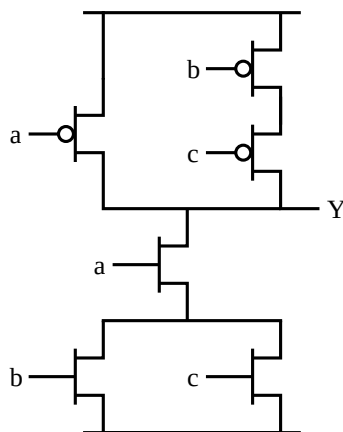
- (a) XOR
- (b) NOR
- (c) XNOR
- (d) NAND

73. For real discrete signal magnitude will be even/odd?_____, Phase will be even/odd?_____

74. plot the output voltage

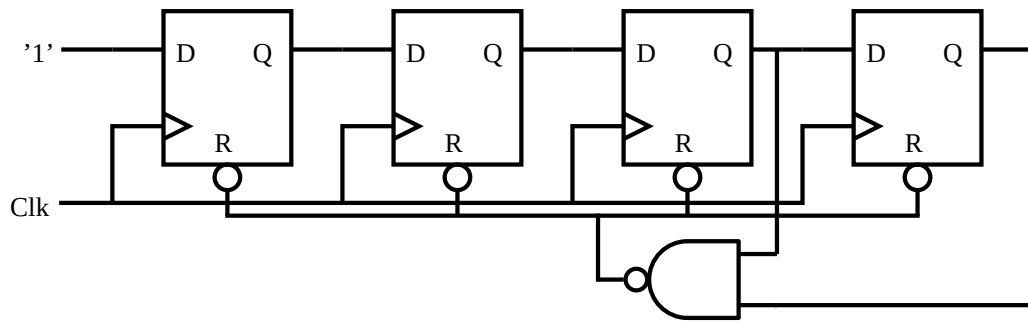


75. Logic of the gate shown is



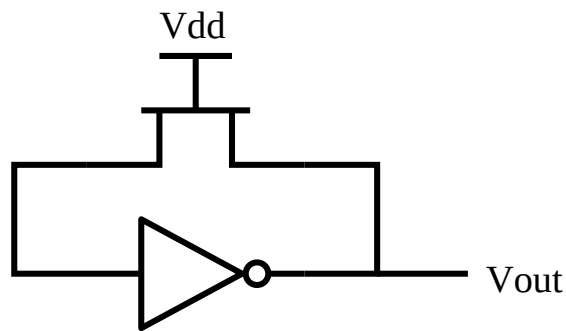
76. The following logic implements a (R is asynchronus reset)

- (a) MOD-15 counter

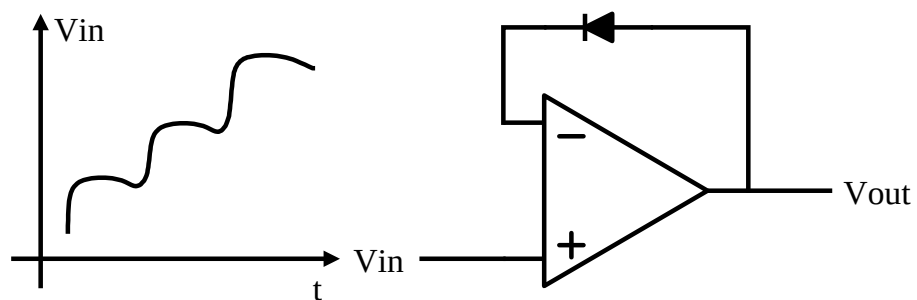


- (b) MOD-8 counter
- (c) MOD-11 counter
- (d) None

77. Determine V_{out} . Give a possible application of the circuit.



78. Plot V_{out} .



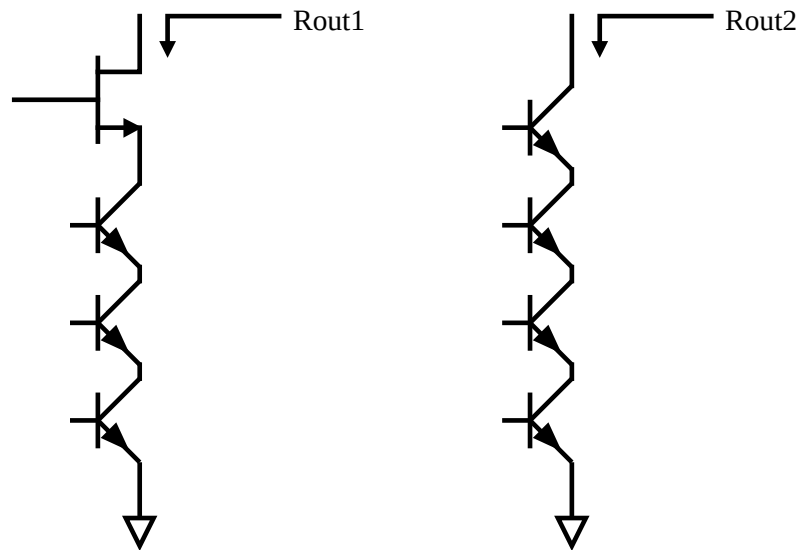
79. What is the equation of FIFO when it is empty?

80. Two clock domains 10MHz and 100MHz. One full period pulse is produced at 5th clock cycle of 100MHz of pulse width equal to $1/100\text{MHz}$. The pulse should be received by 10MHz clock. Rig up the circuit.

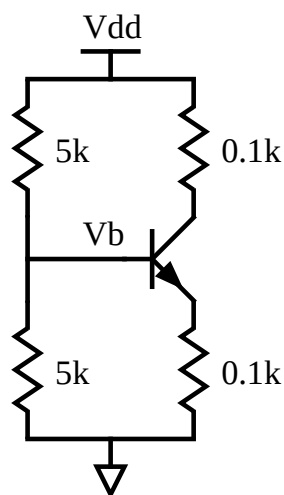
81. There is a set of 5 bits a_5, a_4, a_3, a_2, a_1 . Set for mod 8 counter is

- (a) a_1
- (b) a_2, a_1
- (c) a_3, a_2, a_1
- (d) a_4, a_3, a_2, a_1
- (e) a_5, a_4, a_3, a_2, a_1

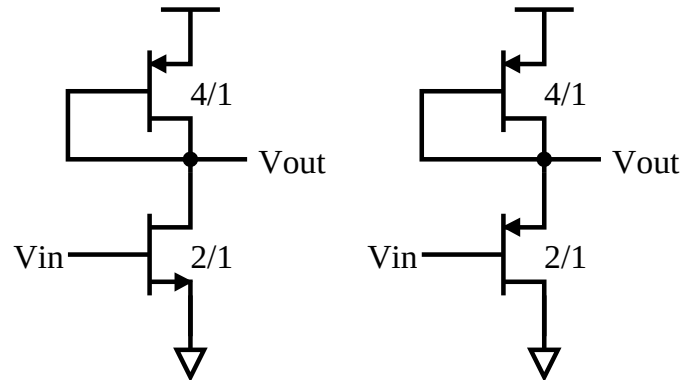
82. Which of the following circuits has higher output impedance?



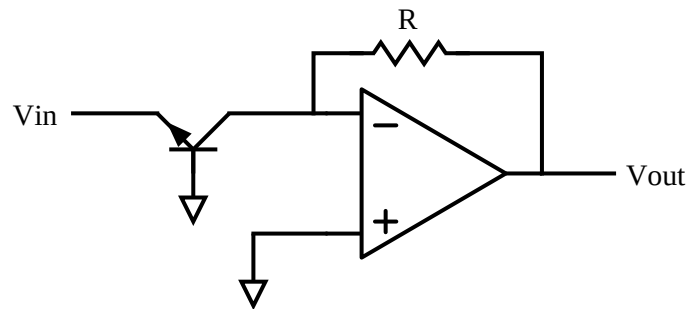
83. Determine i_b, i_c, i_e and V_b with $\beta = 100$ and $V_{be} = 0.7V$ for the circuit below.



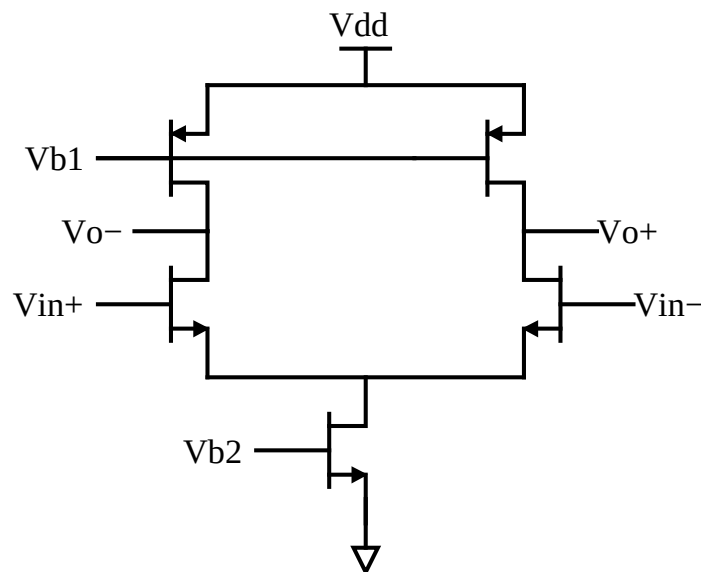
84. Determine gain of following circuit. Assume $\mu_n = 2\mu_p$



85. Determine V_{out}



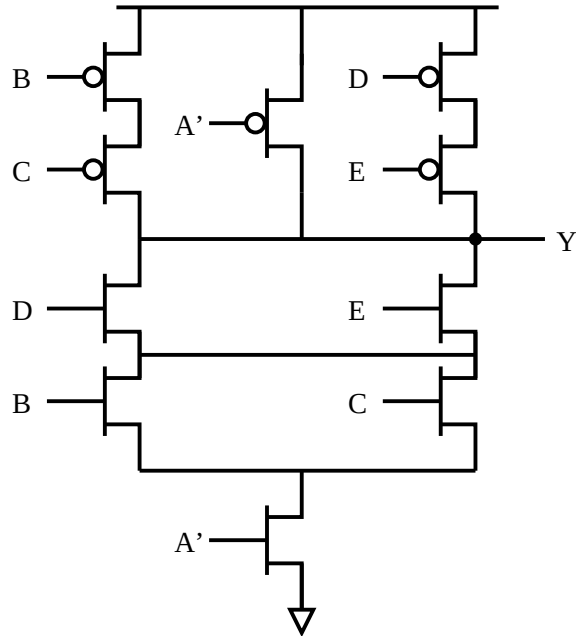
86. Determine the gain for the circuit below.



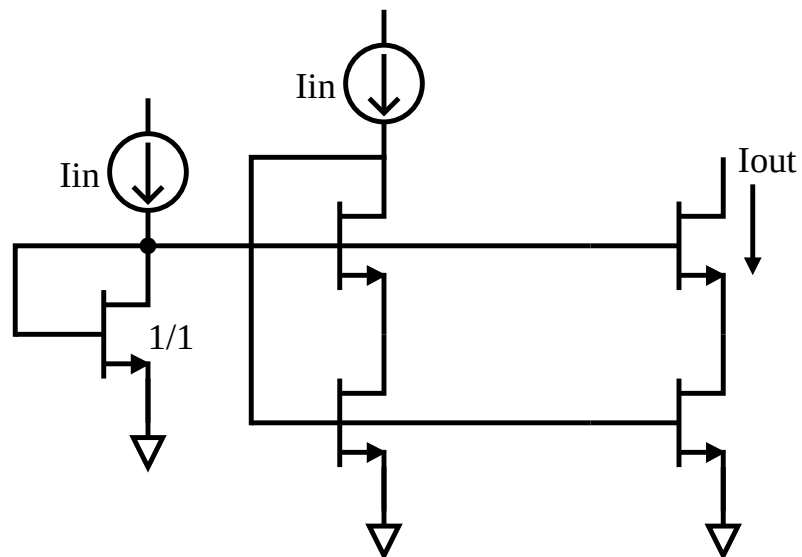
87. Frequency response for an Open loop Opamp was given and the opamp was operated as an inverting amplifier with gain -2. Plot the closed loop frequency response.

88. Number of Boolean equation for an 'n' input variable.

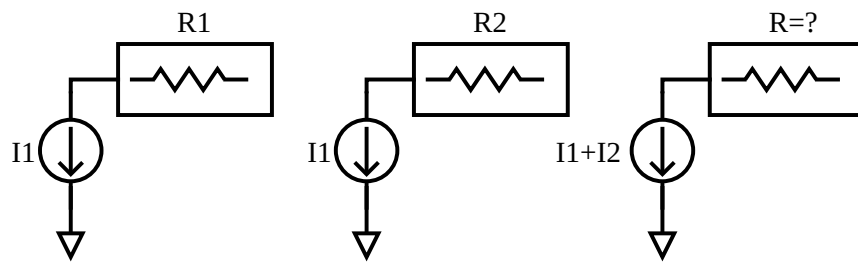
89. Find equation for the circuit below.



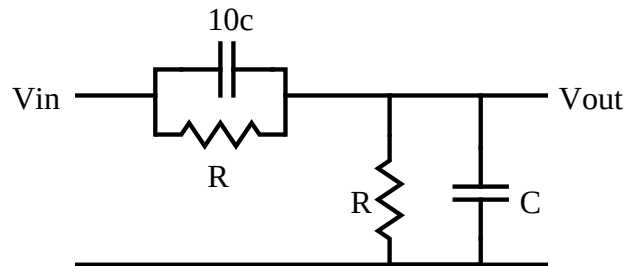
90. Size the remaining 4 transistors for the circuit below so that $I_{out} = I_{in}$.



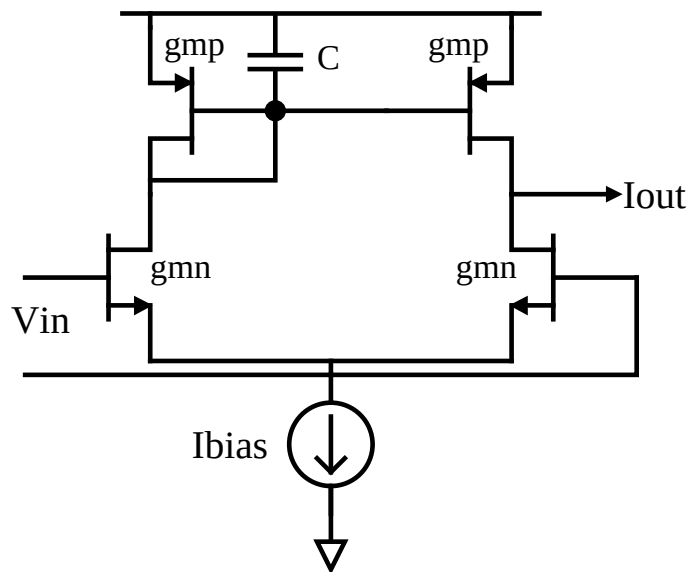
91. Calculate R. The network is fully resistive.



92. What is the time constant



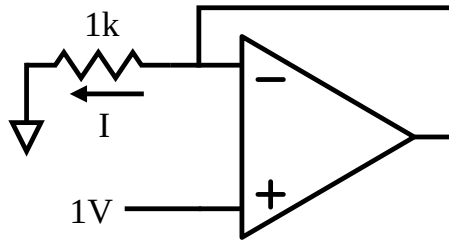
93. Compute the gain at low and high frequency.



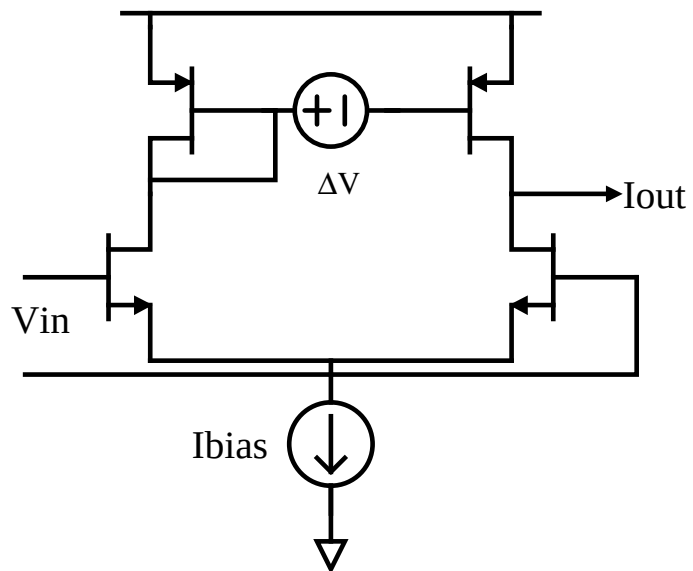
94. The open-loop gain bandwidth product of an opamp is 200 MHz. If it is operating in negative feedback mode with a closed loop gain of 2, then what is the bandwidth?

95. A signal $x(t) = 0.4\sin(2\pi \cdot 75\text{Hz} \cdot t) + 0.6\sin(2\pi \cdot 25\text{Hz} \cdot t)$ is sampled with a sampling frequency of 100 Hz. What will be the frequency spectrum of sampled signal?

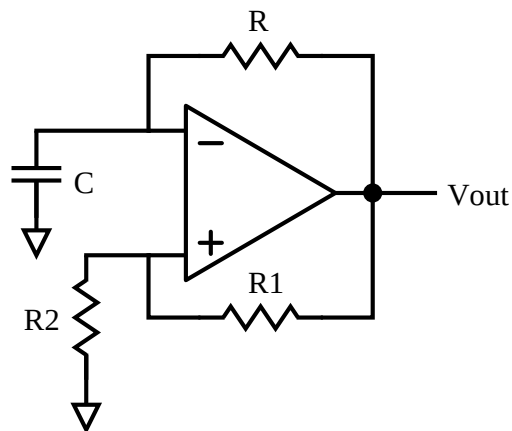
96. Calculate the current I through resistor.



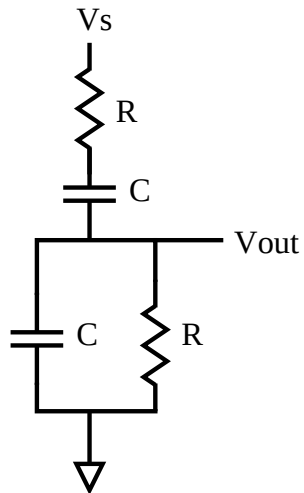
97. Calculate $I_{out}(V_{in}, g_{mn}, g_{mp}, \Delta V)$



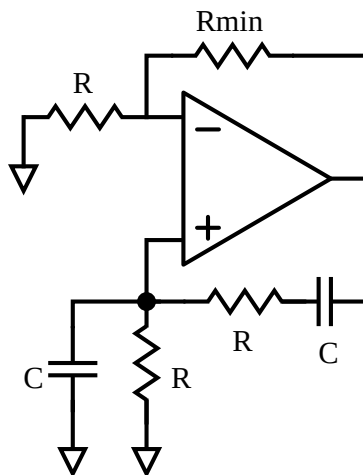
98. What is the frequency of oscillation of output voltage?



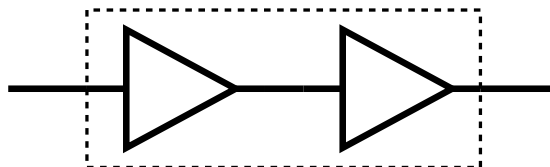
99. Calculate V_{out} .



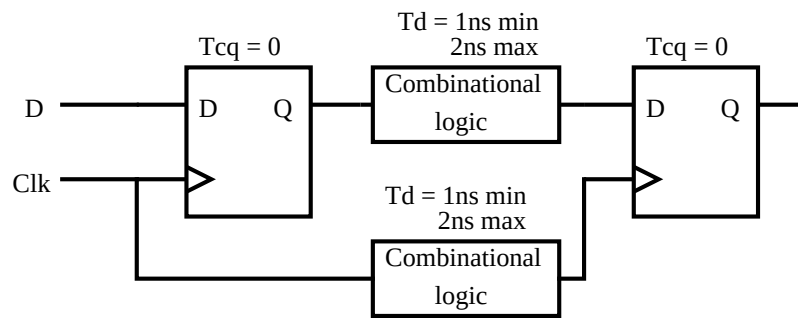
100. What is the minimum value of resistor R_{min} ?



101. Two identical opamps having 3dB frequency f_o are in cascade. Find the overall 3dB frequency.



102. Find the maximum frequency of operation of the following circuit.

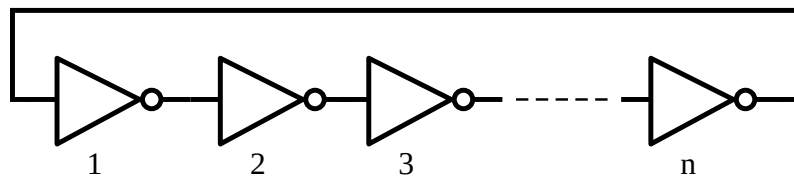


103. Maximum power is dissipated in an inverter when

- (a) Input is 1
- (b) Input is 0
- (c) Input is toggling

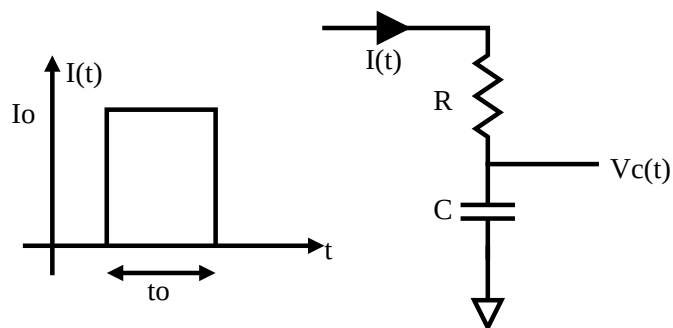
104. In a D Flip-Flop, Q bar is connected to D. Find its function.

105. In a ring oscillator frequency depends on

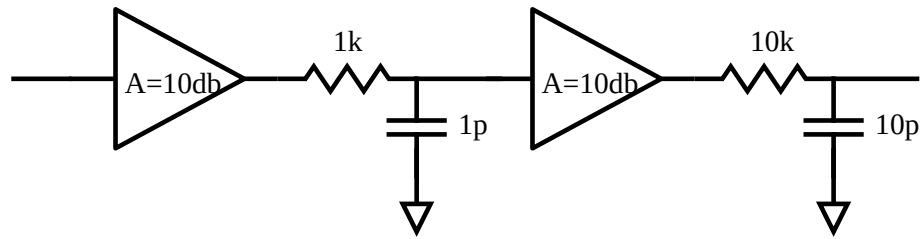


- (a) No of stages n
- (b) Supply voltage
- (c) $\frac{W}{L}$ ratio of the inverters

106. Plot $V_c(t)$.

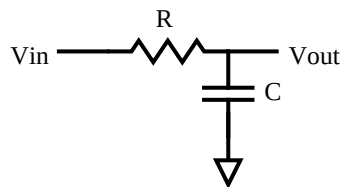


107. Draw the bode plots for gain and phase.

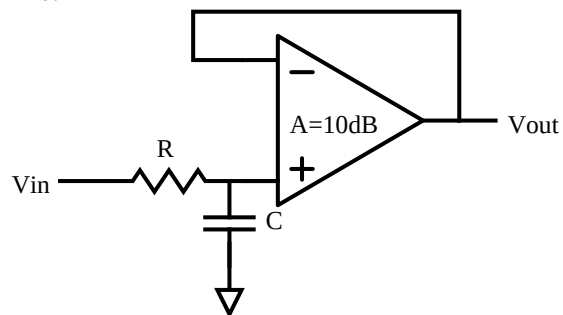


108. Plot the Gate Capacitance versus Gate voltage curves for MOSFETs at DC and high frequencies.

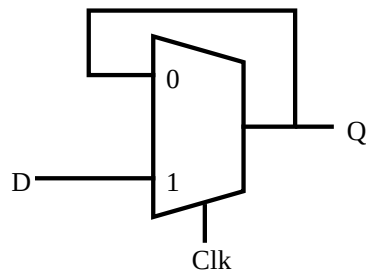
109. Which of these is equivalent to the given RC circuit.



Ans:

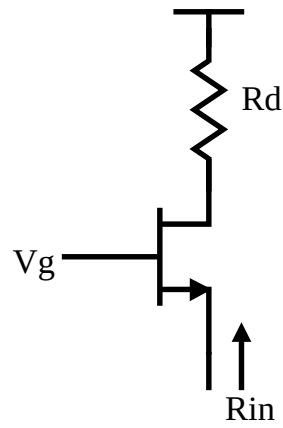


110. The following circuit is a

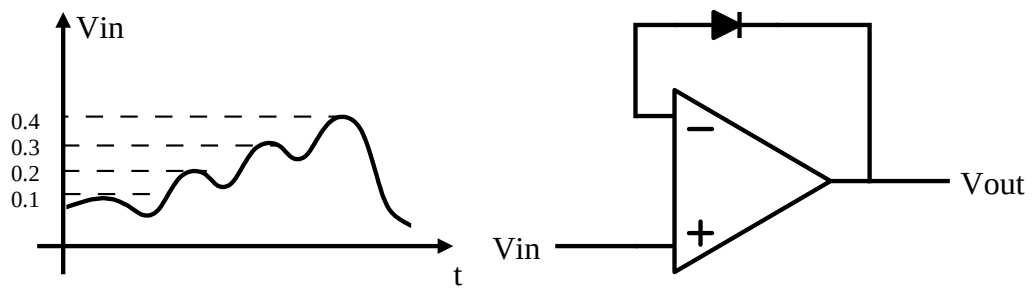


- (a) level 0 transparent latch
- (b) level 1 transparent latch

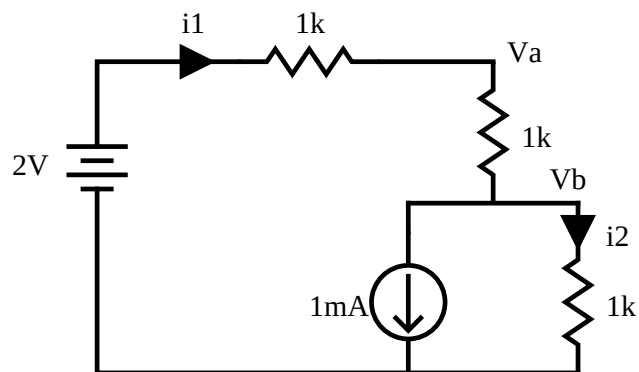
111. Find R_{in} for the following circuit.



112. Find V_{out} .

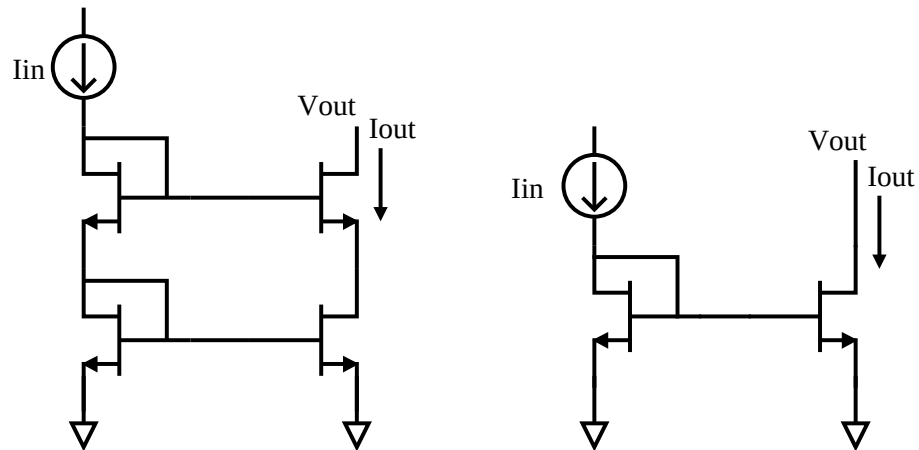


113. Find i_1 , i_2 , V_a and V_b .

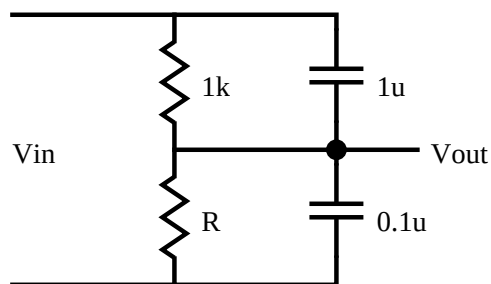


114. Derive the expression for gain of an inverter.

115. Plot I_o Versus V_o for the following circuits.



116. Find R such that V_o/V_{in} is constant.



117. How many distinct boolean expressions are possible with 3 input variables?

- (a) 3
- (b) 8
- (c) 256

118. Plot OUT for the following circuit. What is the function of the given circuit?

